

Thoufiq Yahya

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Senior Embedded Software Engineer | C++ | Qt/QML | Embedded Linux | Automotive HMI

PROFESSIONAL SUMMARY

Senior Embedded Software Engineer and Technical Lead with 13+ years of experience across embedded software, Linux-based platforms, real-time graphics, and user-interface systems. More than 10 years of experience developing production Qt/QML and C++ applications for automotive infotainment and embedded devices, supported by earlier work in set-top box middleware and game development. Strong in software architecture, reusable framework design, performance optimization, system integration, debugging, CI/CD collaboration, and technical leadership. Currently expanding toward embedded systems architecture through focused work in BSP concepts, bare-metal fundamentals, and Android Open Source Project (AOSP) internals.

CORE COMPETENCIES

Embedded Software and Architecture

- Modern C++ (C++11/14/17), object-oriented design, SOLID principles, design patterns
- Qt Framework, Qt Quick, QML, JavaScript, reusable UI frameworks, component architecture
- Embedded Linux, Linux application development, system integration, IPC, D-Bus
- Software architecture, UML, technical design, code reviews, design documentation
- Requirements analysis, feature delivery, defect reproduction, root-cause analysis

Performance and Graphics

- OpenGL ES, GLSL, HLSL, Qt Rendering Framework, GPU-accelerated rendering
- Real-time graphics, 3D visualization, shader development, rendering pipelines
- CPU, memory, and rendering performance analysis on resource-constrained hardware
- Blender and GLTF asset workflows, Unity feasibility evaluation

Development and Delivery

- Python, Bash, JavaScript, Kotlin, Rust, ActionScript, PHP
- CMake, GCC, cross-compilation, Docker, build configuration
- Git, CI/CD collaboration, GDB, JIRA, TDD, GTest, GMock
- Agile software development, technical mentoring, cross-functional collaboration

Architecture Development

- BSP and board-level software concepts
- Bare-metal programming fundamentals
- Android Open Source Project (AOSP) architecture and platform fundamentals
- Embedded system design, boot flow, hardware-software interfaces, platform integration

PROFESSIONAL EXPERIENCE

Xpheno Staffing | Embedded Software Engineer (Client: MBRDI)

December 2024 - Present

- Own and enhance an in-house real-time 3D visualization engine used in automotive infotainment experiences.
- Develop Qt-based visualization features for interactive 3D vehicle models, user interactions, and feature demonstrations.
- Implement GPU-accelerated rendering effects and custom visualizations using GLSL and the Qt Graphics and Rendering Framework.
- Optimize rendering performance, shader execution, memory usage, and graphics workflows for resource-constrained embedded Linux hardware.
- Investigate runtime behavior, reproduce defects, and resolve performance and integration issues with platform and graphics teams.
- Collaborate with UX, graphics, platform, and integration stakeholders to define requirements and deliver production features.
- Evaluate Unity-based rendering architecture and contribute to technical direction for software-defined vehicle platforms.

Mercedes-Benz Research & Development India | Senior Software Engineer / Technical Lead

April 2016 - September 2023

- Led a five-engineer team responsible for architecture, development, code reviews, mentoring, and delivery of production infotainment widgets.
- Designed reusable C++, Qt6, and QML frameworks and UI components for embedded Linux automotive platforms.
- Built advanced controls and interaction patterns, including animated components, progress indicators, and shader-driven effects.
- Applied SOLID principles, object-oriented design, and design patterns to create maintainable and extensible software.
- Translated product and UX requirements into technical designs, UML models, and production-ready implementations.
- Coordinated with platform, CI/CD, release, integration, and testing teams across India and Germany throughout the product lifecycle.
- Diagnosed integration, runtime, and resource-usage issues using Linux development tools, GDB, Git, and GCC-based workflows.
- Led an Ubuntu 16.04-to-20.04 development environment migration using Bash scripts and Docker-based workflows.
- Applied test-driven development practices with GTest and GMock to improve software quality.

- Designed a modular Sound Dialogue System prototype with a context-aware dialogue manager and integrations with Dialogflow, Rasa, Wit.ai, and Houndify.

L&T Technology Services | Technical Consultant (Client: Cisco NDS)**June 2014 - April 2016**

- Developed embedded television middleware and Electronic Program Guide (EPG) applications for set-top box platforms.
- Built UI components using QML, JavaScript, ActionScript, and Adobe Flash during the migration to Qt/QML-based embedded applications.
- Collaborated with Linux platform and hardware teams on system integration, build configuration, and application troubleshooting.
- Optimized UI responsiveness, rendering performance, and memory usage on resource-constrained embedded devices.

I2Play Interactive | Game Developer**November 2012 - June 2014**

- Developed multiplayer educational games and interactive learning systems using ActionScript and Flash.
- Built gameplay systems, animations, visual effects, and interactive components for browser-based applications.
- Integrated SmartFoxServer for real-time multiplayer functionality and developed supporting PHP/MySQL services.
- Contributed to game-engine development using object-oriented programming and design patterns.

Excel Goodies | Game Developer / Web Designer**June 2011 - January 2012**

- Developed educational games and rich internet applications using ActionScript, Flash, PHP, HTML, JavaScript, and CSS.
- Created reusable UI components, game assets, mockups, and visual designs for client projects.
- Delivered end-to-end solutions in a startup environment across design, development, and client collaboration.

AWARDS

- Star Performer, L&T Technology Services, 2015
- Winner, Innovation Boot Camp, Mercedes-Benz R&D, 2019
- Silver Star, Mercedes-Benz R&D, 2020

EDUCATION

Bachelor of Information Technology**2006 - 2009**

Sadakathullah Appa College, Manonmaniam Sundaranar University, Tirunelveli, India